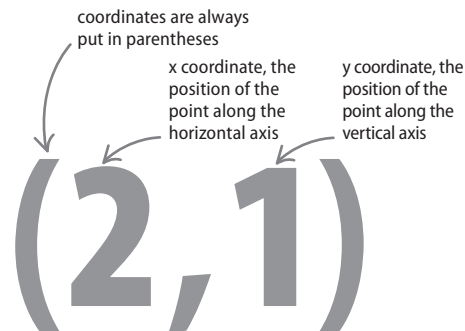
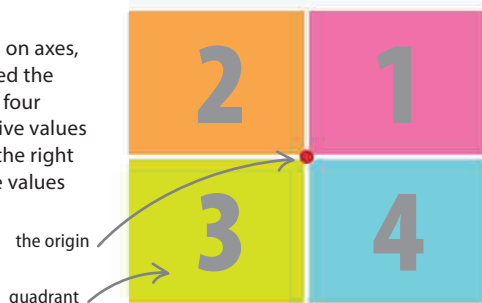


Graph coordinates

Coordinates are used to identify the positions of points on graphs, in relation to two axes—the y axis is a vertical line, and the x axis is a horizontal line. The coordinates of a point are written as its position on the x axis, followed by its position on the y axis, (x, y).

► Four quadrants

Coordinates are measured on axes, which cross at a point called the “origin.” These axes create four quadrants. There are positive values on the axes above and to the right of the origin, and negative values below and to its left.

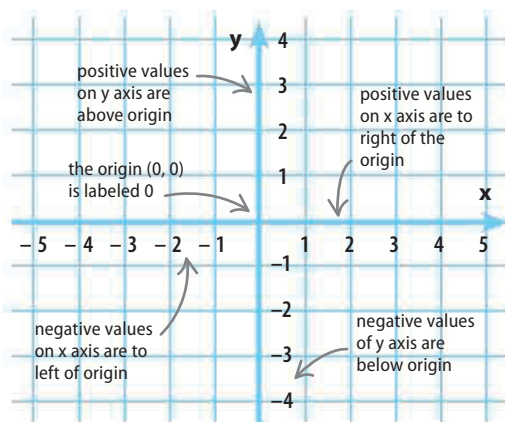


△ Coordinates of a point

Coordinates give the position of a point on each axis. The first number gives its position on the x axis, the second its position on the y axis.

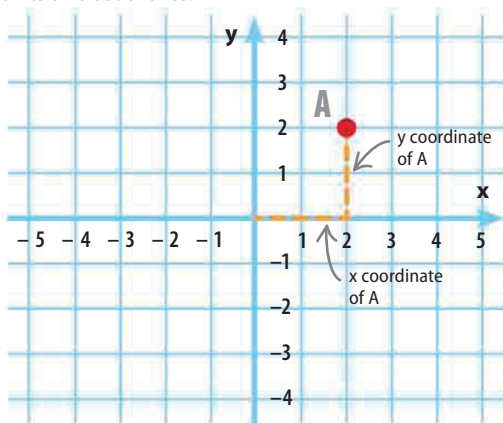
Plotting coordinates

Coordinates are plotted on a set of axes. To plot a given point, first read along to its value on the x axis, then read up or down to its value on the y axis. The point is plotted where the two values cross each other.



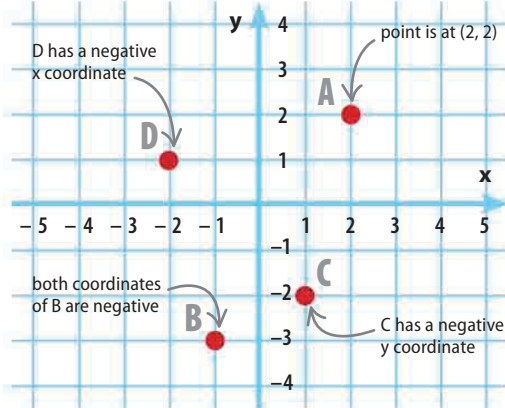
$$A = (2, 2) \quad B = (-1, -3) \\ C = (1, -2) \quad D = (-2, 1)$$

These are four sets of coordinates. Each gives its x value first, followed by its y value. Plot the points on a set of axes.



To plot each point, look at its x coordinate (the first number), and read along the x axis from 0 to this number. Then read up or down to its y coordinate (the second number).

Using graph paper, draw a horizontal line to form the x axis, and a vertical line for the y axis. Number the axes, with the origin separating the positive and negative values.



Plot each point in the same way. With negative coordinates, the process is the same, but read to the left instead of right for an x coordinate, and down instead of up for a y coordinate.